

Supported iron Fischer-Tropsch catalyst: superior activity and stability using a thermally stable silica-doped alumina support

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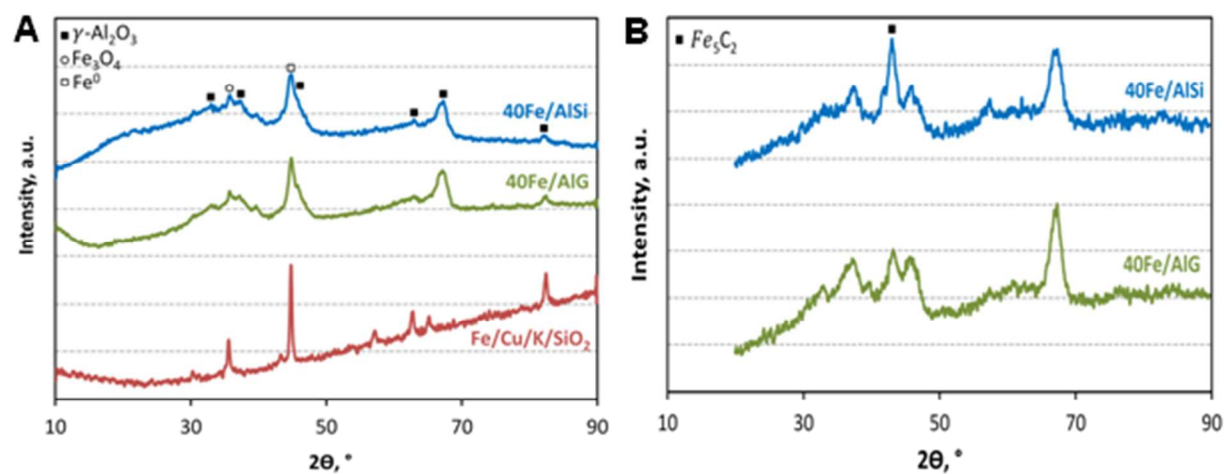


Figure S 1. X-ray diffraction patterns of (a) reduced and (b) carbided catalysts

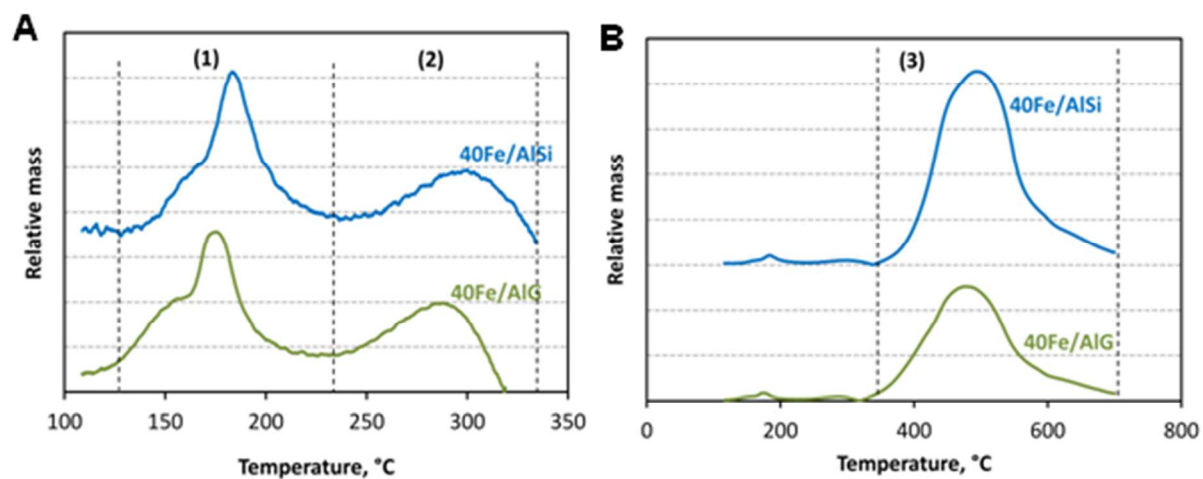


Figure S 2. Syngas-TPR profiles of the 40Fe/AlSi and 40Fe/AlG catalysts (a) up to 350 °C and (b) up to 700 °C.

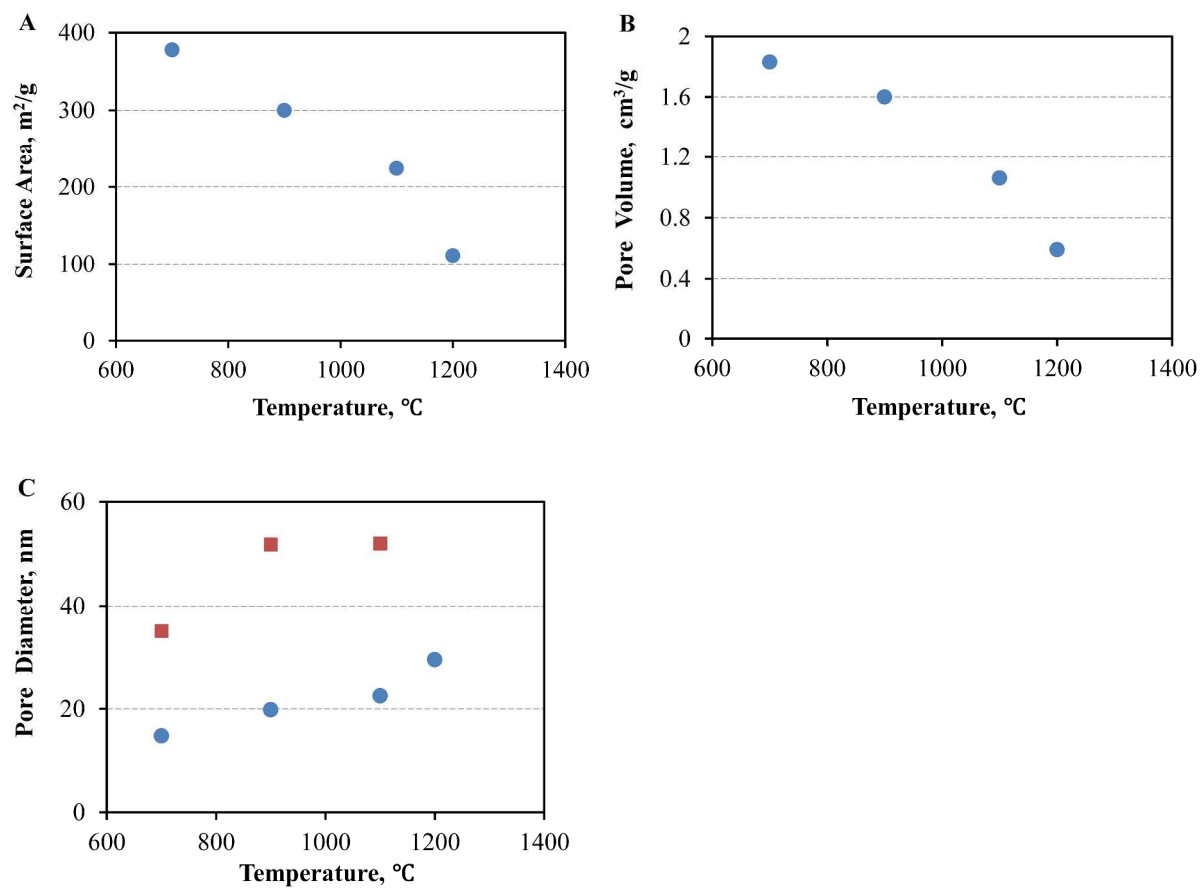


Figure S 3. (a) BET surface area, (b) pore volume, and (c) average pore diameters of AlSi as a function of calcination temperature. The two sets of data for the pore diameter correspond to the two peaks in the bimodal distribution shown in Figure 1 and Table 1 in the text.

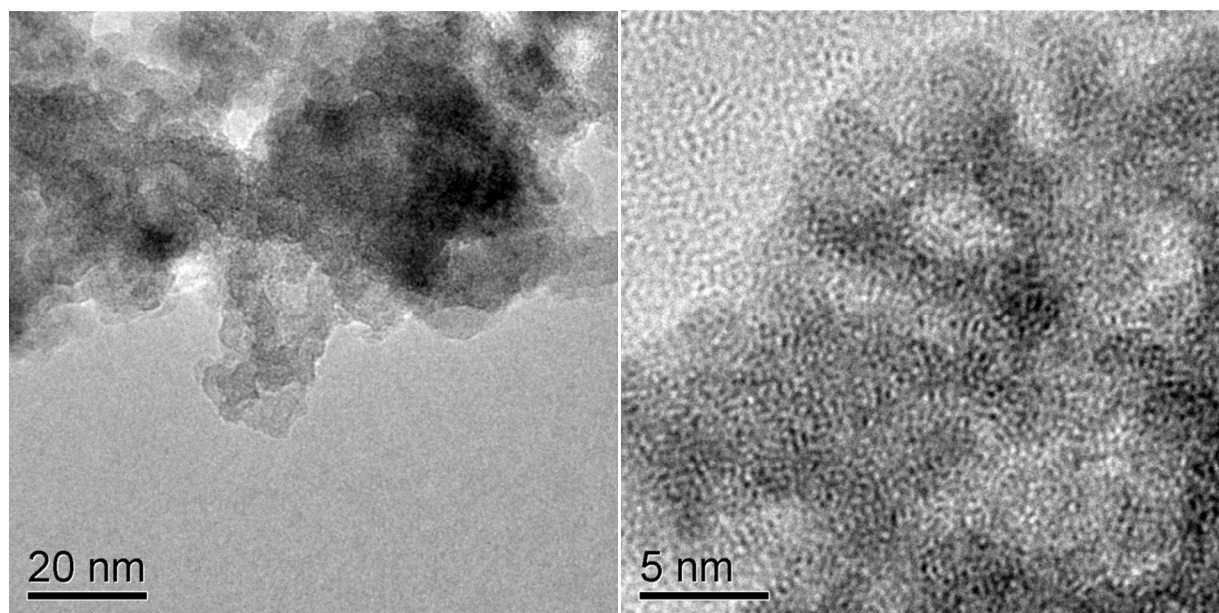


Figure S 4. TEM micrograph of 5% silica doped alumina calcined at 1100°C.

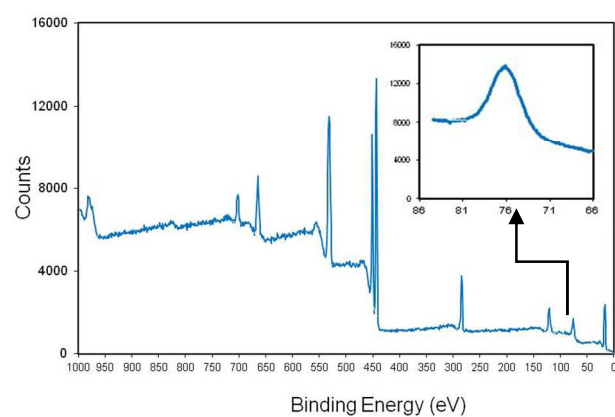


Figure S 5. (A) Al (2p) XPS peak of 5% silica doped alumina, and (B) XPS survey scan of 5% silica doped alumina. This shows an Al (2p) peak at 76.15 eV which is assigned to the alumina structure, thus indicating we have an alumina structure and not mullite.

Table S 1. Comparison of BET properties of different support materials.

Support Material	Si, wt%	Calcination temp, °C	SA (m ² /g)	Vpore (cm ³ /g)	dpore (nm)	XRD phase at
AlSi-this study	5	1100	184	1.1	23, 52	Gamma
AlG-this study	0	700	216	0.65	18.1	Gamma
Sasol-Puralox 1	0		140	0.50	16	
Sasol-Siral 1	5		128	0.40	11	
Horiuchi 2	5	1100	100	1.3	10	Delta

Table S 2. Fe⁰, Fe₃O₄, Fe₅C₂ crystallite sizes estimated from XRD for three catalysts after reduction and carbiding.

Catalyst	XRD, nm		
	Reduced catalysts		Carbided catalysts
	Fe ₃ O ₄	Fe ⁰	Fe ₅ C ₂
40Fe/AlG	6.4	11.6	8.1
40Fe/AlSi	4.4	9.4	7.9
FeCuKSi	20.8	26	-

Table S 3. Performance of three catalysts of this study in fixed-bed reactor activity tests.
T = 260 °C, H₂/CO = 1, P_{H₂}⁰ = P_{CO}⁰ = 0.66 MPa, P_{tot} = 2.1 MPa.

Catalyst ID	40Fe/AlSi *	40Fe/AlG	FeCuKSi
Space velocity, NI/g _{cat} /h	22.0	22.3	11.4
Time on stream, h	246	126	246
CO conversion, %	20	19	32
(H ₂ +CO) Conversion, %	20.7	17.3	28
<u>Activity</u>			
<i>Rate</i> mmol(CO+H ₂)/g _{cat} /h	153	107	91
<i>Rate constant</i> mmol(CO+H ₂)/g _{Fe} /h/MPa	618	429	215
<u>Selectivity</u>, mol%			
CO ₂	38.1	42.7	46.7
CH ₄	9.4	10.0	4.0
C ₃ +	47.0	43.0	46.5
<u>Catalyst productivity</u>			
g _{HC} /g _{cat} /h	0.45	0.39	0.34

*This test was conducted at 256 °C, the activity was corrected to 260 °C using E_A = 100 kJ/mol.

Table S 4. Comparison of BYU 40Fe/AlSi with other *supported* Fe catalysts at 260 °C.

	BYU-this study ¹	TAMU-Bukur	TAMU-Bukur	U. Kentucky-Davis ²	BYU-Bartholomew
Catalyst	40Fe/AlSi ³	Fe/SiO ₂ ⁴	Fe/Al ₂ O ₃ ⁵	Fe/Al ₂ O ₃ ⁶	10Fe/SiO ₂ ⁷
Literature reference	This study	³	³	⁴	⁵
Reactor type	Fixed-bed	Slurry	Slurry	Slurry	Fixed-bed
Pressure, MPa	2.1	1.5	1.5	1.3	1.0
Inlet molar H ₂ /CO	0.66	0.67	0.67	0.7	1.0
Space velocity, NI/g _{cat} /h	5.6	1.35	1.32	0.68	1.92
TOS, h	512	100	100	300	150
CO conversion, %	72	--	--	60	77
%CO converted to CO ₂	46	--	--	--	47
rate, mmol(CO)/g _{cat} /h	77	--	--	28	12 ²
rate, mmol(CO+H ₂)/g _{cat} /h	133	36	19	--	17 ²
k _{cat} , mmol(CO+H ₂)/g _{cat} /MPa/h	325	101	51	--	50 ²
k _{Fe} , mmol(CO+H ₂)/g _{Fe} /MPa/h	878	300	150	--	504 ²
H.C. selectivities, mol%					
CH ₄	9.6	7.0	3.5	5.8 ⁸	6.8
C ₃₊	86.0	--	--	--	--
C ₁ +C ₂	12.4	--	--	--	--
Cat. Prod., g _{HC} /g _{cat} /h	0.50	--	--	0.12	0.3
Cat. Prod., g _{HC} /g _{Fe} /h	1.25	--	--	0.53	3

¹ P_{H₂}⁰ = 0.6 MPa, P_{CO}⁰ = 0.9 MPa, P_{tot} = 2.1 MPa

² rates are corrected to 260 °C using E_A = 100 kJ/mol

³ 100Fe/7.5Cu/4K/150Al₂O₃.

⁴ 100Fe/5Cu/6K/139SiO₂.

⁵ 100Fe/5Cu/9K/139Al₂O₃.

⁶ 100Fe/6Cu/8K/250Al₂O₃.

⁷ 100Fe/10Pt/2K/888SiO₂.

⁸ at 250°C

Table S 5. Comparison of BYU 40Fe/AlSi with *unsupported* Fe catalysts at 260°C

	BYU-this study ¹	TAMU-Bukur	Kentucky-Davis	Berkeley-Iglesia	Ruhrchemie	Mobil
Catalyst	40Fe/AlSi ³	Unsupp. ⁴	Unsupp. ⁵	Unsupp. ⁶	Unsupp. ⁷	Unsupp.
Literature reference	This study	⁶	⁷	⁸	⁹	¹⁰
Reactor type	Fixed-bed	slurry	slurry	Fixed-bed	slurry	slurry
Run temperature, °C	260	260	260	235	260	257
Pressure, MPa	2.1	2.2	1.30	2.14	1.5	1.5
Inlet molar H ₂ /CO	0.66	0.68	0.67	2.0	0.67	0.73
Space velocity, NI/g _{cat} /h	5.6	3.5		11.0	2.2	2.3 ⁸
TOS	512	314			86	475
CO conversion, %	72	84	79	51	66	90
rate, mmol(CO)/g _{cat} /h	77	76			38	34 ²
k _{cat} , mmol(CO)/g _{cat} /MPa/h	188	170		185 ²	97	
rate, mmol(CO+H ₂)/g _{cat} /h	133	120	168		61	49 ²
k _{cat} , mmol(CO+H ₂)/g _{cat} /MPa/h	325	269			155	
k _{Fe} , mmol(CO+H ₂)/g _{Fe} /MPa/h	878	450			290	265 ²
H.C. selectivities, mol%						
CH ₄	9.6	2.6			5.3	2.3
C ₃₊	86.0	90.1				92
C ₁ +C ₂	12.4	6.3				5
Cat. Prod., g _{HC} /g _{cat} /h	0.50	0.51	0.74		0.27	
Cat. Prod., g _{HC} /g _{Fe} /h	1.25	0.86	1.22		0.51	0.44 ²

¹ P_{H₂}⁰ = 0.6 MPa, P_{CO}⁰ = 0.9 MPa, P_{tot} = 2.1 MPa

² rates are corrected to 260 °C using E_A = 100 kJ/mol

³ 100Fe/7.5Cu/4K/150Al₂O₃.

⁴ 100Fe/3Cu/4K/16SiO₂.

⁵ 100Fe/5.1Si/2Cu/3.0K.

⁶ Fe/Zn/K/Cu.

⁷ 100Fe/5Cu/4.2K/25SiO₂.

⁸ NI/g_{Fe}/h

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